

Wireless Communication Numerical Companion

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

Ch 5 • Equalization and Diversity Numerical Problems & Solutions

No PYQ in 22 papers has set a pure calculation on this chapter. What is examined, repeatedly, is the **MSE optimum-weight derivation** **HOT 3/22**, which the papers ask for as bookwork with a diagram. The rest of this companion works the tutorial-set problems, because they are the standard follow-ups and they make the diversity theory concrete.

The formula sheet

- **MSE cost function** $\xi = E[x_k^2] + \omega^T \mathbf{R} \omega - 2\mathbf{p}^T \omega$
- **Optimum weights** $\hat{\omega} = \mathbf{R}^{-1} \mathbf{p}$, $\xi_{\min} = E[x_k^2] - \mathbf{p}^T \hat{\omega}$
- **LMS update** $\omega_{k+1} = \omega_k + \mu e_k \mathbf{y}_k^*$
- **Selection diversity**, M Rayleigh branches, mean SNR Γ each:

$$\Pr[\text{SNR} \leq \gamma] = \left(1 - e^{-\gamma/\Gamma}\right)^M, \quad \frac{\bar{\gamma}_M}{\Gamma} = \sum_{k=1}^M \frac{1}{k}$$

- **Maximal ratio combining** $\gamma_M = \sum_{i=1}^M \gamma_i$, so $\bar{\gamma}_M = M \Gamma$
- **Interleaver** degree m , word length n : an (n, k) code handling bursts of $b < (n-k)/2$ becomes an (mn, mk) code handling bursts of mb

Problem 1 Optimum tap weights for a 2-tap equalizer

HOT 3/22 79 Bh 8, 79 Ch 7, 80 Ch 7 · the papers ask for the **derivation**; this worked example shows the derivation actually being used

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8] · 2079 Chaitra · Q6 · [7] · 2080 Chaitra · Q6 · [7]

Determine the optimal solution of the weights using MSE algorithm with its appropriate diagram and derivation.

PROBLEM not a PYQ · numbers chosen to exercise the result **verified/added**

A 2-tap adaptive equalizer sees an input correlation matrix $\mathbf{R} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$, a cross-correlation vector $\mathbf{p} = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$ and a transmitted signal power $E[x_k^2] = 1$. Find the optimum weight vector $\mathbf{w}_{opt}$ and the minimum mean square error ξ_{min} .

Step 1: why $\hat{\omega} = \mathbf{R}^{-1} \mathbf{p}$ (the derivation)

- Error signal, with $\mathbf{y}_k$ the input vector and x_k the known training symbol:

$$e_k = x_k - \mathbf{y}_k^T \omega$$

- Square it and take expectations:

$$\xi = E[e_k^2] = E[x_k^2] + \omega^T \mathbf{R} \omega - 2\mathbf{p}^T \omega$$

with $\mathbf{R} = E[\mathbf{y}_k \mathbf{y}_k^T]$ and $\mathbf{p} = E[x_k \mathbf{y}_k]$.

- ξ is **quadratic** in ω , so the surface is a bowl with **one** minimum. Set the gradient to zero:

$$\nabla \xi = 2\mathbf{R}\omega - 2\mathbf{p} = 0 \Rightarrow \boxed{\hat{\omega} = \mathbf{R}^{-1} \mathbf{p}}$$

Step 2: invert $\mathbf{R}$

$$\det \mathbf{R} = (1)(1) - (0.5)(0.5) = 0.75$$

$$\mathbf{R}^{-1} = \frac{1}{0.75} \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1.3333 & -0.6667 \\ -0.6667 & 1.3333 \end{bmatrix}$$

$\mathbf{R}$ must be non-singular for the solution to exist. Here $\det \mathbf{R} = 0.75 \neq 0$ ✓

Step 3: the optimum weights

$$\hat{\mathbf{w}} = \begin{bmatrix} 1.3333 & -0.6667 \\ -0.6667 & 1.3333 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$$

$$w_0 = 1.3333(0.8) - 0.6667(0.5) = 1.0667 - 0.3333 = 0.7333$$

$$w_1 = -0.6667(0.8) + 1.3333(0.5) = -0.5333 + 0.6667 = 0.1333$$

$$\hat{\mathbf{w}} = \begin{bmatrix} 0.7333 \\ 0.1333 \end{bmatrix}$$

Step 4: the minimum MSE

$$\begin{aligned} \xi_{\min} &= E[x_k^2] - \mathbf{p}^T \hat{\mathbf{w}} = 1 - [(0.8)(0.7333) + (0.5)(0.1333)] \\ &= 1 - (0.5867 + 0.0667) = 1 - 0.6533 = 0.3467 \end{aligned}$$

- **Interpretation:** the equalizer has removed about 65 % of the error energy. The residual 0.3467 is what no linear 2-tap filter can remove.
- **Adding taps lowers** $\xi_{\min}$, but at the cost of processing time and slower convergence. That trade-off is the practical answer to “how many taps?”.
- **The examiner wants the derivation, not the arithmetic.** Reproduce Steps 1 and 2 symbolically and show the bowl-shaped error surface; the numbers are the easy part.

Problem 2 Selection diversity: probability of a deep fade

tutorial set, and the standard companion to the “why implement diversity” question **TOP 11/22**

PROBLEM not a PYQ · the standard companion to the “why diversity” question **verified/added**

Five branches of a selection diversity receiver each receive an independent Rayleigh fading signal. The average SNR on every branch is 20 dB. Determine the probability that the SNR of the selected branch drops below 10 dB, compute the mean SNR after selection, and compare both with a single receiver having no diversity.

“Why is there a need to implement diversity?” is asked in 11 of the 22 papers; this is the arithmetic that answers it with a number. **verified/added**

Step 1: form the ratio γ/Γ (in linear terms, not dB)

- Threshold $\gamma = 10 \text{ dB} = 10^{10/10} = 10$
- Mean per branch $\Gamma = 20 \text{ dB} = 10^{20/10} = 100$
-

$$\frac{\gamma}{\Gamma} = \frac{10}{100} = 0.1$$

Subtracting in dB gives the same answer, but watch the order: $10 \text{ dB} - 20 \text{ dB} = -10 \text{ dB}$, and $10^{-10/10} = 0.1$. ✓ Either route is fine; mixing them is not.

Step 2: probability the SNR falls below the threshold

$$\Pr[\text{SNR} \leq \gamma] = \left(1 - e^{-\gamma/\Gamma}\right)^M = \left(1 - e^{-0.1}\right)^M = (0.09516)^M$$

Branches M	$\Pr[\text{SNR} \leq 10 \text{ dB}]$	Meaning
1 (no diversity)	9.516×10^{-2}	the link is below threshold 9.5 % of the time
2	9.056×10^{-3}	about 1 in 110
4	8.201×10^{-5}	about 1 in 12 000
5	7.804×10^{-6}	about 1 in 128 000

- **Five-branch selection diversity improves the outage probability by more than four orders of magnitude** over a single receiver, from 9.5×10^{-2} to 7.8×10^{-6} .
- That is the entire justification for diversity, expressed as a number.

Step 3: mean SNR after selection combining

$$\frac{\bar{\gamma}_M}{\Gamma} = \sum_{k=1}^M \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = 2.2833$$

$$\bar{\gamma}_5 = 100 \times 2.2833 = 228.33 = \boxed{23.59 \text{ dB}}$$

- So the mean SNR rises only **3.59 dB**, from 20 dB to 23.59 dB.
- **The important point:** the gain in **mean SNR** is modest, but the gain in **outage probability** is enormous. Diversity works by cutting the **tail** of the distribution, not by lifting the average. **Say this in the theory answer too.**
- **Diminishing returns:** $M=2$ gives 1.5Γ (1.76 dB), $M=3$ gives 1.833Γ (2.63 dB), $M=4$ gives 2.083Γ (3.19 dB), $M=5$ gives 2.283Γ (3.59 dB). **Most of the benefit is already in the second branch**, which is why practical systems use 2, occasionally 4, and almost never 5.

Step 4: compare with maximal ratio combining

- MRC output SNR is the **sum** of the branch SNRs, so

$$\bar{\gamma}_M = M\Gamma = 5 \times 100 = 500 = \boxed{26.99 \text{ dB}}$$

- MRC gains a full **6.99 dB** against selection diversity's 3.59 dB, and its outage probability at 10 dB falls to 7.67×10^{-8} , another factor of 100 better.
- **The price** is M full receivers plus accurate estimation of each branch's amplitude and phase. Selection diversity needs only to compare SNRs.

Problem 3 Interleaver design against burst errors

HOT 3/22 80 Bh, 81 Bh, 81 Ba · the papers ask for the **principle**; this shows the design sum behind it

QUESTION AS PRINTED 2080 Bhadra · Q5 (b) · [4] · 2081 Bhadra · Q7 · [4] · 2081 Baishakh · Q5 · [2+6]

Why interleaving is needed in wireless communication? Explain its working in brief. · Elaborate the working principle behind Interleaving with an appropriate diagram. · Define Interleaving.

PROBLEM not a PYQ · the design sum behind the principle **verified/added**

An interleaver has m rows of n -bit words. Each word carries k source bits and $(n - k)$ parity bits of a block code. Show how the interleaver and coder together break up a burst of channel errors, and size the interleaver for a $(15, 11)$ block code with interleaver degree $m = 8$: give the block size, the source bits per block, and the longest burst the arrangement can correct.

Step 1: what the block code alone can do

- A block code with minimum distance d' corrects t errors when $d' \geq 2t + 1$.
- For an (n, k) code the parity budget is $n - k$, so it can handle a burst of

$$b < \frac{n - k}{2}$$

- For the $(15, 11)$ code: $n - k = 4$, so $t = 2$ and a burst of up to **2 bits** per codeword is correctable.

Step 2: what the interleaver does

- Coded bits are written into the array **by column** and read out **by row**.
- Two bits that were adjacent in a codeword end up m **bit periods apart** on the channel.
- So a channel burst of length $l = mb$ is spread across m different codewords, giving each of them only b errors, which each can correct.

Step 3: the resulting combined code

- An (n, k) code plus a degree- m interleaver behaves as an

$$(mn, mk) \text{ block code}$$

- For $(15, 11)$ with $m = 8$:

$$(8 \times 15, 8 \times 11) = \boxed{(120, 88) \text{ block code}}$$

- Burst-handling capability:

$$l = m \times b = 8 \times 2 = \boxed{16 \text{ consecutive bits}}$$

- **The code rate is unchanged** at $88/120 = 11/15 = 0.733$. The interleaver costs no redundancy at all: it buys burst protection purely with **delay and memory**.

Step 4: the delay penalty

- The interleaver must fill all $mn = 120$ bits before the first row can be sent, and the de-interleaver must fill 120 bits before the first row can be decoded.
- At a coded rate of, say, 22.8 kbps (GSM full rate) that is

$$\frac{2 \times 120}{22\,800} = 10.5 \text{ ms of added delay}$$

- **Human speech tolerates delays below about 40 ms**, which is the hard cap on how deeply a voice system may interleave. Data services, with no such limit, interleave far more deeply.

Problem 4 Choosing the equalizer algorithm from the channel

the reasoning behind "explain any two adaptive equalization algorithms" (73 Ma, 74 Ma)

QUESTION AS PRINTED 2073 Magh · Q5 (a) · [4] · 2074 Magh · Q10 (a) · [5]

Explain briefly adaptive equalization algorithms (any two). · Write short note on Adaptive Equalization.

PROBLEM not a PYQ · the reasoning that picks the algorithm verified/added

A GSM mobile travels at 70 km/h on a 900 MHz carrier through a channel whose rms delay spread is $\sigma_\tau = 5 \mu\text{s}$. The symbol rate is 270.833 ksym/s. Decide how many equalizer taps are needed, how often the equalizer must retrain, and hence which adaptive algorithm (LMS, RLS or Kalman) suits the link.

- **Step 1: how many taps?** The tap span must cover the **maximum expected delay spread**, in symbol periods.

$$T_s = \frac{1}{270\,833} = 3.692 \mu\text{s}$$

$$\frac{\sigma_\tau}{T_s} = \frac{5}{3.692} = 1.35 \text{ symbols}$$

Allowing for the maximum excess delay being several times σ_τ , an equalizer spanning **about 5 symbol periods** is needed.

GSM equalizers are specified to handle 4 to 5 symbols of delay spread, which matches.

- **Step 2: how fast does the channel change?**

$$v = 70 \text{ km/h} = 19.44 \text{ m/s}, \quad \lambda = 0.3333 \text{ m}, \quad f_m = \frac{v}{\lambda} = 58.33 \text{ Hz}$$

$$T_c \approx \frac{0.423}{f_m} = \frac{0.423}{58.33} = 7.25 \text{ ms}$$

- **Step 3: retraining rate.** The equalizer must converge and be used well inside 7.25 ms.

$$\text{minimum retrain rate} = \frac{1}{7.25 \text{ ms}} = 138 \text{ times per second}$$

GSM sends a training midamble in every 4.615 ms frame, ie 217 times per second, comfortably inside the requirement.

✓

- **Step 4: which algorithm?**

- The channel changes every few milliseconds, and the GSM training sequence is only **26 bits** long.
- **LMS** typically needs hundreds of iterations to converge, so 26 bits is not enough.
- **RLS** converges in roughly $2N$ iterations for an N -tap filter, ie about 10 for a 5-tap equalizer. **RLS, or an MLSE with a channel estimator, is the only viable choice.**
- That is exactly why GSM receivers use **MLSE with Viterbi**, not a simple LMS transversal equalizer.

Problem 5 Cell-edge SNR gain from two-branch diversity

the arithmetic behind “how diversity improves the quality of the network” (76 Bh 2)

QUESTION AS PRINTED 2076 Bhadra · Q5 · [2+4]

Explain how diversity improves the quality of network. Explain the operation of RAKE receiver with appropriate block diagram.

PROBLEM not a PYQ · the number that answers “how much” verified/added

On a Rayleigh channel the mean SNR at the cell edge is 15 dB and the receiver needs 8 dB to demodulate. Compare the outage probability with a single antenna and with two-branch selection diversity, and state the coverage gain this implies.

– **Step 1: the ratio.**

$$\gamma = 10^{8/10} = 6.31, \quad \Gamma = 10^{15/10} = 31.62, \quad \frac{\gamma}{\Gamma} = 0.1995$$

– **Step 2: outage with one antenna.**

$$\text{Pr} = 1 - e^{-0.1995} = 0.1809 = \boxed{18.1 \%}$$

– **Step 3: outage with two branches.**

$$\text{Pr} = (0.1809)^2 = 0.03273 = \boxed{3.27 \%}$$

– **Step 4: what that is worth in dB.** To reach 3.27 % outage with a *single* antenna the mean SNR would have to satisfy $1 - e^{-\gamma/\Gamma'} = 0.03273$, ie $\gamma/\Gamma' = 0.03327$, giving

$$\Gamma' = \frac{6.31}{0.03328} = 189.6 = 22.78 \text{ dB}$$

$$\text{diversity gain} = 22.78 - 15 = \boxed{7.78 \text{ dB}}$$

– **Interpretation:** a second antenna is worth almost 8 dB of transmit power at this operating point, for no extra spectrum and no extra power. **That is why every base station has two receive antennas.**

– **Coverage implication.** With a path loss exponent $n = 3.5$, 7.78 dB of margin extends the cell radius by

$$10^{7.78/(10 \times 3.5)} = 10^{0.2223} = 1.67 \times$$

ie **about 2.8 times the area from the same site.**

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB,	double, single, vestigial sideband
		VSF	
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	BT	GMSK filter bandwidth $\times$ symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
WLAN	wireless local area network	WMAN	wireless metropolitan area network
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
5G	5th generation mobile communication network	5G	5th generation mobile communication network

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.